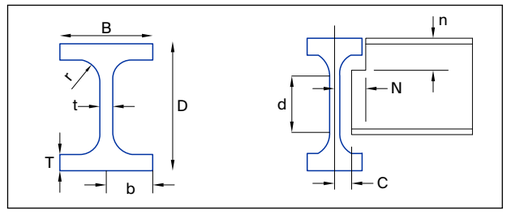
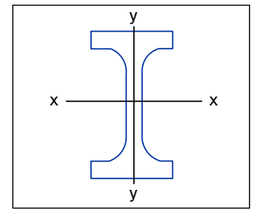
2.2
Wide Flange Beams (HE)



Dimensions	EN10365
Specification	EN10025
Size Range	HE 100 AA to HE 1000 x 584



Section designation	Dimensions													Properties												
	Mass per Metre	Depth of Section	Width of Section	Thickness		Root	Depth between Fillets	Area of Section	Ratios for Local Buckling		Dimensions for Detailing			Surface Area	Second Moment of Area		Radius of Gyration		Elastic Modulus		Plastic Modulus		Buckling Parameter	Torsional Index	Warping Constant	Torsional Constant
				Flange	Web				Flange	Web	End clearance	Notch														
		D	B	T	t	r	d	A	b/T	d/t	C	N	n	Per metre	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	u	x	H	J
	kg/m	mm	mm	mm	mm	mm	mm	cm²			mm	mm	mm	m²	cm⁴	cm⁴	cm	cm	cm³	cm³	cm³	cm³			dm⁶	cm⁴
HE 800 A	224	790.0	300.0	28.0	15.0	30.0	674.0	286	44.9	4.02	10	154	58	2.70	303,000	12,600	32.6	6.65	7,680	843	8,700	1,310	7.799	29.5	18.3	609
HE 800 B	262	800.0	300.0	33.0	17.5	30.0	674.0	334	38.5	3.37	11	152	64	2.71	359,000	14,900	32.8	6.68	8,980	994	10,200	1,550	7.813	25.6	21.8	959
HE 800 M	317	814.0	303.0	40.0	21.0	30.0	674.0	404	32.1	2.78	13	152	70	2.75	443,000	18,600	33.1	6.79	10,900	1,230	12,500	1,930	7.876	21.6	27.8	1,660
HE 800 x 373	373	826.0	308.0	46.0	25.0	30.0	674.0	475	27.0	2.42	15	152	76	2.78	524,000	22,500	33.2	6.89	12,700	1,460	14,700	2,310	7.927	19.0	34.1	2,550
HE 800 x 444	444	842.0	313.0	54.0	30.0	30.0	674.0	566	22.5	2.06	17	152	84	2.82	634,000	27,800	33.5	7.01	15,100	1,780	17,600	2,830	7.987	16.4	42.8	4,160
HE 900 AA	198	870.0	300.0	20.0	15.0	30.0	770.0	252	51.3	5.63	10	154	50	2.86	301,000	9,040	34.6	5.99	6,920	603	8,000	958	7.357	42.5	16.3	322
HE 900 A	252	890.0	300.0	30.0	16.0	30.0	770.0	320	48.1	3.73	10	152	60	2.90	422,000	13,600	36.3	6.50	9,480	903	10,800	1,410	7.711	31.8	25.0	749
HE 900 B	291	900.0	300.0	35.0	18.5	30.0	770.0	371	41.6	3.16	11	152	66	2.91	494,000	15,800	36.5	6.53	11,000	1,050	12,600	1,660	7.723	27.7	29.5	1,150
HE 900 M	333	910.0	302.0	40.0	21.0	30.0	770.0	424	36.7	2.76	13	152	70	2.93	570,000	18,400	36.7	6.60	12,500	1,220	14,400	1,930	7.766	24.6	34.8	1,680
HE 900 x 391	391	922.0	307.0	46.0	25.0	30.0	770.0	498	30.8	2.41	15	152	76	2.97	674,000	22,300	36.8	6.70	14,600	1,450	17,000	2,310	7.815	21.6	42.6	2,600
HE 900 x 466	466	938.0	312.0	54.0	30.0	30.0	770.0	594	25.7	2.06	17	152	84	3.01	815,000	27,600	37.1	6.81	17,400	1,770	20,400	2,830	7.877	18.7	53.4	4,230
HE 1000 AA	222	970.0	300.0	21.0	16.0	30.0	868.0	282	54.3	5.33	10	152	52	3.06	406,000	9,500	38.0	5.80	8,380	633	9,780	1,020	7.235	45.7	21.3	387
HE 1000 x 249	249	980.0	300.0	26.0	16.5	30.0	868.0	317	52.6	4.30	10	152	56	3.08	481,000	11,800	39.0	6.09	9,820	784	11,400	1,240	7.437	39.7	26.6	582
HE 1000 A	272	990.0	300.0	31.0	16.5	30.0	868.0	347	52.6	3.60	10	152	62	3.10	554,000	14,000	40.0	6.35	11,200	934	12,800	1,470	7.618	34.9	32.1	835
HE 1000 B	314	1,000.0	300.0	36.0	19.0	30.0	868.0	400	45.7	3.07	12	152	66	3.11	645,000	16,300	40.2	6.38	12,900	1,080	14,900	1,720	7.634	30.6	37.6	1,270
HE 1000 M	349	1,008.0	302.0	40.0	21.0	30.0	868.0	444	41.3	2.76	13	152	70	3.13	722,000	18,500	40.3	6.45	14,300	1,220	16,600	1,940	7.673	27.8	43.0	1,710
HE 1000 x 393	393	1,016.0	303.0	43.9	24.4	30.0	868.0	500	35.6	2.49	14	150	74	3.14	808,000	20,500	40.2	6.40	15,900	1,350	18,500	2,170	7.635	25.4	48.1	2,330
HE 1000 x 415	415	1,020.0	304.0	46.0	26.0	30.0	868.0	529	33.4	2.37	15	150	76	3.15	853,000	21,700	40.2	6.41	16,700	1,430	19,600	2,300	7.635	24.3	51.1	2,700
HE 1000 x 438	438	1,026.0	305.0	49.0	26.9	30.0	868.0	556	32.3	2.23	15	150	80	3.17	909,000	23,400	40.4	6.48	17,700	1,530	20,800	2,460	7.684	23.0	55.3	3,180
HE 1000 x 494	494	1,036.0	309.0	54.0	31.0	30.0	868.0	629	28.0	2.02	18	150	84	3.19	1,028,000	26,800	40.4	6.53	19,800	1,740	23,400	2,820	7.702	20.9	64.0	4,400
HE 1000 x 584	584	1,056.0	314.0	64.0	36.0	30.0	868.0	744	24.1	1.70	20	150	94	3.24	1,246,000	33,400	40.9	6.70	23,600	2,130	28,000	3,480	7.805	18.0	81.2	7,150